

Comparative Analysis of Bandgap Reference Circuits Using Different Amplifier Topologies

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Abstract—For precise operation of analog and mixed signal ICs, a temperature independent reference voltage source is necessary. In this paper, we discuss the design of two types of band gap reference circuits using the diode. The circuits are designed and analyzed in a 180 nm CMOS technology using the CADENCE Virtuoso tool. Both designs use forward biased diodes to form CTAT and PTAT voltages, which are later added together to form a temperature independent reference voltage source.

The main difference in design of the circuits is the type of feedback amplifier used; in one design, a two-stage NMOS differential amplifier is used, while in another design, a folded cascode amplifier is used. The performance of the designed circuits is studied through simulations under different operating conditions such as variations in temperature, changes in power supply voltage, and frequency response.

The findings show that the BGR circuit with folded cascode amplifier has better temperature independence, improved PSRR, and improved line and load regulation because of high gain and high output impedance. On the other hand, the differential amplifier-based BGR design shows low power consumption and simple circuit design.

Index Terms—Bandgap reference (BGR), CMOS analog circuits, diode-based reference, differential amplifier, folded cascode amplifier, temperature coefficient, PSRR, line regulation.

I. INTRODUCTION

Voltage reference circuits are one of the essential components in the design of analog and mixed-signal integrated circuits. Analog-to-digital converters (ADCs), digital-to-analog converters (DACs), power management units, phase-locked loops (PLLs), sensor interfaces, and System-on-Chip (SoC) subsystems all need a reference voltage that is stable and accurate. The accuracy of the reference has a direct impact on the system's linearity, noise margin, and overall performance. A voltage reference should not change when the temperature, supply voltage, process changes, or load conditions change.

The bandgap reference (BGR) circuit, which Widlar first came up with in 1971 and Brokaw and Kuijk later improved, is now the standard way to make reference voltages that don't change with temperature in both CMOS and bipolar technologies. The basic idea behind the BGR is to cancel out two temperature-dependent quantities that have opposite properties:

- **CTAT (Complementary-to-Absolute-Temperature):** The base-emitter voltage (V_{BE}) of a forward-biased p-n

junction decreases with temperature at approximately $-2 \text{ mV}/^\circ\text{C}$.

- **PTAT (Proportional-to-Absolute-Temperature):** The thermal voltage ($V_T = kT/q$) increases linearly with absolute temperature at approximately $+0.085 \text{ mV}/^\circ\text{C}$.

When these two parts are properly scaled and added together, the voltage comes out to be about 1.205 eV at 0 K, which is close to the extrapolated silicon bandgap energy. This gives a reference voltage of about 1.2 V with a first-order temperature coefficient that is almost zero.

Parasitic bipolar transistors, diode-connected MOSFETs, or forward-biased p-n junction diodes can be used to make CTAT and PTAT parts in CMOS technology. The diode-based method is especially good for standard digital CMOS processes that don't make high-quality bipolar devices because it uses the built-in source-bulk or drain-bulk junctions of MOSFETs.

The feedback operational amplifier in a BGR circuit ensures accurate summation of PTAT and CTAT components by enforcing equality conditions at the summing node, thereby minimizing offset-induced temperature drift.

The type of feedback amplifier you choose has a big effect on important performance factors like temperature coefficient (TC), power supply rejection ratio (PSRR), line regulation, and noise. The standard two-stage NMOS differential amplifier has a moderate open-loop gain (usually 60–80 dB in 180 nm CMOS), is easy to design, and can easily be compensated for frequency. But its low gain and PSRR can make reference accuracy worse when the temperature and supply voltage change. The folded cascode operational amplifier, on the other hand, has a higher open-loop gain (usually between 80 and 100 dB) and a better PSRR because it has a high output impedance. However, this makes the circuit more complicated and uses more power.

This paper presents the design, simulation, and comparative analysis of two BGR circuits implemented in the GPDK 180 nm CMOS process using Cadence Virtuoso: (1) a BGR employing a two-stage NMOS differential amplifier, and (2) a BGR utilizing a folded cascode amplifier. In both designs, forward-biased diodes are used to generate PTAT and CTAT components. The temperature coefficient (TC), PSRR, line

regulation, load regulation, and transient response are used to compare the two topologies.

II. LITERATURE REVIEW

A. Evolution of Temperature Compensation Techniques

The bandgap reference (BGR) concept is based on combining two voltages with opposite temperature coefficients to achieve a temperature-independent reference. The base-emitter voltage (V_{BE}) of a bipolar junction transistor, or the forward voltage (V_F) of a diode, exhibits a Complementary-to-Absolute Temperature (CTAT) characteristic, decreasing approximately linearly with temperature at a rate close to $-2\text{ mV}/^\circ\text{C}$.

In contrast, a Proportional-to-Absolute Temperature (PTAT) voltage is generated from the difference in voltages of two junctions operating at different current densities. By properly scaling and summing the CTAT and PTAT components, a reference voltage with minimal temperature dependence can be realized.

In standard 180 nm CMOS technology, first-order compensation techniques usually achieve temperature coefficients (TC) between 20 and 50 ppm/ $^\circ\text{C}$. Modern precision systems require TC values below 10 ppm/ $^\circ\text{C}$.

B. Feedback Topologies: Differential and Folded Cascode

The error amplifier is an essential component in BGR circuits for achieving balance between the PTAT and CTAT branches.

1) *Two-Stage NMOS Differential Amplifier*: The traditional two-stage NMOS differential amplifier enjoys popularity because of its ease of design and relatively large output swing. Nevertheless, it is characterized by average gain (60–70 dB in 180 nm CMOS) and poor PSRR, resulting in offset error problems that impair temperature stability.

2) *Folded Cascode Amplifier*: Folded cascode architecture offers improved gain (over 80 dB) and output resistance in a single stage, hence low output error and a more accurate reference voltage.

Moreover, folded cascode amplifiers are better at rejecting power supply noise, making them ideal for use in current-generation integrated circuits. The major limitation is that they are harder to design than two-stage NMOS differential amplifiers and offer a smaller output swing.

C. Diode-Based Reference Generation in Standard CMOS

Traditional bandgap reference circuits employed specific transistors fabricated in the CMOS process. However, currently, diode-based configurations are gaining popularity due to their small size and low voltage operation.

In the 180 nm General Purpose Design Kit (GPDK) technology, these diode-based circuits can be implemented in an area-efficient manner and have predictable behavior with temperature if the value of the diode ideality factor n is known.

However, the design imposes stricter requirements on the selection of the amplifier, making it essential to select an appropriate amplifier.

D. Summary of Technical Gap

A lot of research has been done on bandgap reference design but not many studies directly compare the standard NMOS differential amplifier with the folded cascode amplifier inside a diode-based bandgap circuit using 180 nm GPDK technology.

This work fills that gap by comparing both designs in terms of temperature stability, power supply rejection, and line regulation using Cadence Virtuoso simulations.

III. METHODOLOGY

The proposed bandgap reference (BGR) design is based on the principle of generating a temperature-independent reference voltage by combining two complementary temperature-dependent voltages: Complementary-to-Absolute Temperature (CTAT) and Proportional-to-Absolute Temperature (PTAT). The CTAT voltage is derived from the forward voltage of a diode, which decreases with temperature, while the PTAT voltage is obtained from the difference between two diode voltages operating at different current densities.

A. BGR Principle and Analytical Modeling

The output reference voltage is given by:

$$V_{ref} = V_{BE} + k \cdot \Delta V_{BE}$$

where V_{BE} represents the CTAT component, ΔV_{BE} represents the PTAT component, and k is a scaling factor determined by the resistor network.

The PTAT voltage is expressed as:

$$\Delta V_{BE} = V_T \ln(n)$$

where $V_T = \frac{kT}{q}$ is the thermal voltage and n is the ratio of current densities in the two diode devices. Proper selection of k ensures first-order temperature compensation, resulting in a nearly constant reference voltage over temperature variations.

B. Circuit Topology Selection

Two amplifier topologies are used in this work to investigate their impact on BGR performance:

- **Two-Stage NMOS Differential Amplifier**: A simple and widely used architecture offering ease of design and wider output swing, but with moderate gain and poor PSRR performance.
- **Folded Cascode Amplifier**: A high-gain architecture with superior output impedance and improved PSRR, though with higher design complexity and power consumption.

The objective of using these two topologies is to evaluate their influence on key performance parameters such as temperature stability, supply sensitivity, and overall efficiency.

C. Design Implementation

The BGR circuits were implemented using GPDK 180 nm CMOS technology in Cadence Virtuoso. Both architectures employ diode-connected devices to generate CTAT and PTAT voltages. A resistor network is used to appropriately scale and combine these components to generate a stable reference voltage (V_{ref}).

The operational amplifier in each topology enforces feedback, ensuring proper balance between PTAT and CTAT branches and maintaining a stable operating point.

Bias currents are generated using current mirror circuits, and transistor sizing is optimized to ensure saturation operation, adequate gain, and reduced mismatch effects.

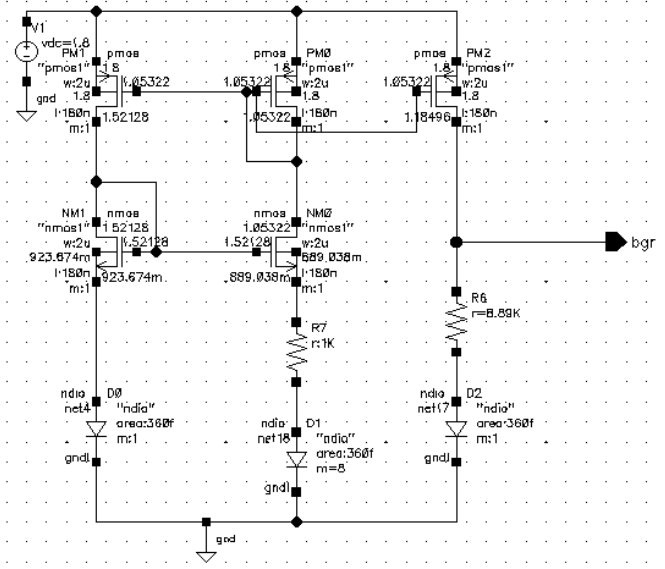


Fig. 1. Bandgap reference circuit using two-stage NMOS differential amplifier.

D. Simulation Strategy

To evaluate performance, both designs were simulated under identical operating conditions using Spectre simulator. The following analyses were performed:

- **Temperature Sweep:** -40°C to 125°C to evaluate temperature stability of V_{ref} .
- **Line Regulation:** Supply voltage variation from 1.8 V to 3.3 V to study sensitivity to supply changes.
- **PSRR Analysis:** AC analysis with small-signal supply perturbation to evaluate noise rejection capability.
- **Load Regulation:** Variation of load current to analyze output voltage stability under different loading conditions.
- **Power Consumption:** Measured using DC operating point analysis.

Power consumption is calculated as:

$$P = V_{DD} \times I_{DD}$$

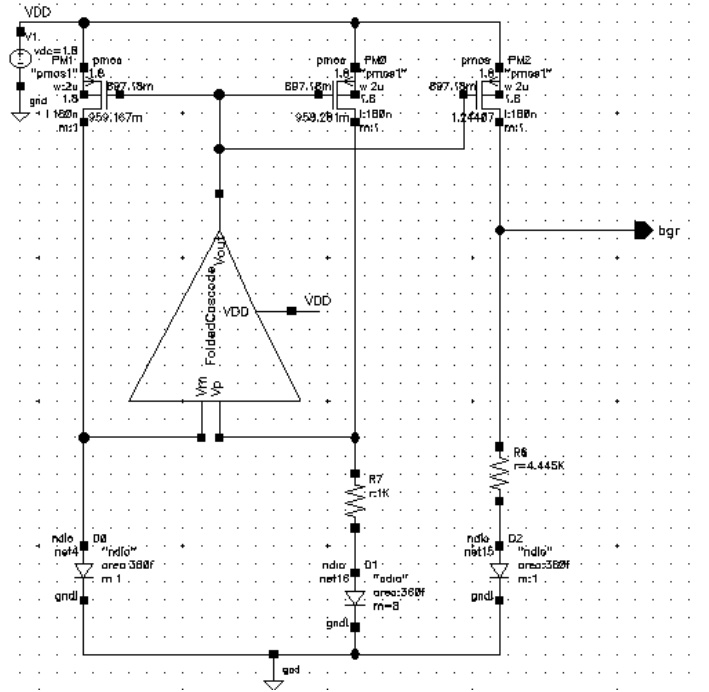


Fig. 2. Bandgap reference circuit using folded cascode amplifier.

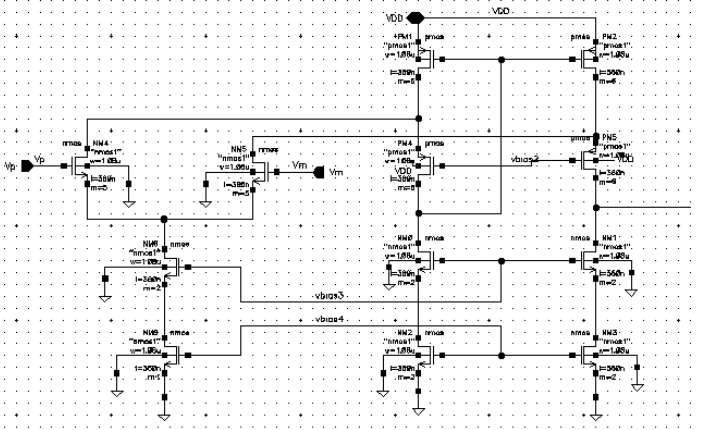


Fig. 3. Folded cascode operational amplifier used in the BGR design.

E. Performance Evaluation Metrics

The performance of both BGR architectures is evaluated using the following key metrics:

- Temperature Coefficient (TC)
- Line Regulation
- Power Consumption
- Power Supply Rejection Ratio (PSRR)
- Load Regulation

These parameters are extracted from simulation results and used to perform a comparative analysis of both amplifier topologies in terms of stability, accuracy, and power efficiency.

IV. RESULTS AND DISCUSSION

The performance of the two bandgap reference (BGR) topologies—two-stage NMOS differential amplifier and folded cascode amplifier—is evaluated using Cadence Virtuoso simulations. For each performance metric, both designs are analyzed separately and compared.

A. Temperature Analysis

The temperature variation of the reference voltage (V_{ref}) was analyzed from -50°C to 125°C .

$$TC = \frac{1}{V_{ref}} \cdot \frac{dV_{ref}}{dT}$$

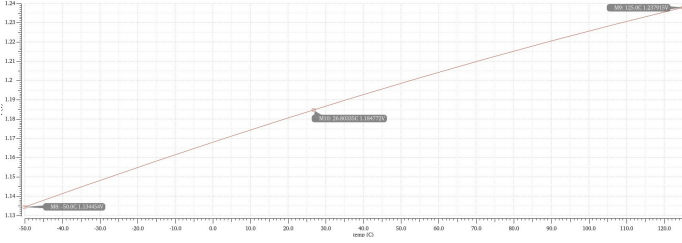


Fig. 4. Temperature variation of V_{ref} for differential amplifier-based BGR.

$$TC \approx 498 \text{ ppm}/^{\circ}\text{C}$$

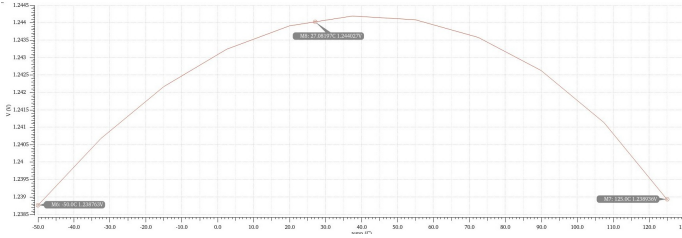


Fig. 5. Temperature variation of V_{ref} for folded cascode-based BGR.

$$TC \approx 0.80 \text{ ppm}/^{\circ}\text{C}$$

The folded cascode-based BGR shows significantly improved temperature stability with a flatter response over the entire range. This is due to higher loop gain, which improves CTAT and PTAT cancellation accuracy. The differential amplifier exhibits higher variation due to limited gain and offset effects.

B. Line Regulation Analysis

The supply voltage was swept from 1.8 V to 3.3 V.

$$\text{Line Regulation} = \frac{\Delta V_{ref}}{\Delta V_{DD}}$$

Line Regulation $\approx 490 \text{ mV/V}$, of Differential Pair

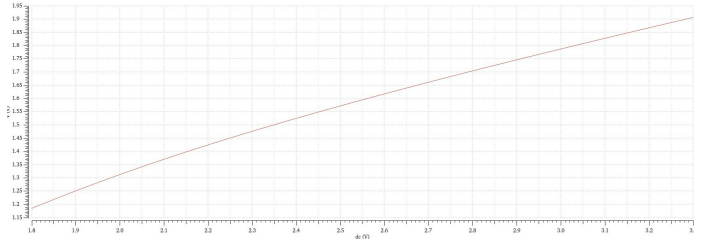


Fig. 6. Line regulation for differential amplifier-based BGR.

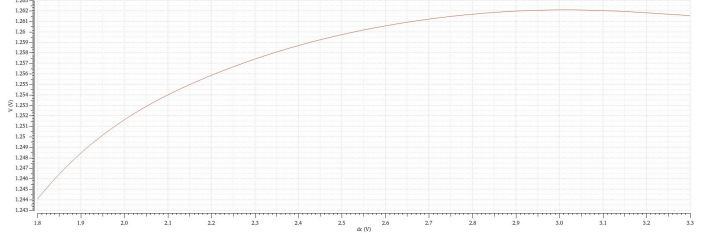


Fig. 7. Line regulation for folded cascode-based BGR.

Line Regulation $\approx 11.7 \text{ mV/V}$, of Folded Cascode

The folded cascode architecture demonstrates better line regulation due to higher output impedance and improved supply rejection. The differential design shows noticeable variation with supply voltage changes.

C. Power Consumption

The power consumption was calculated from the DC operating point by measuring the supply current (I_{DD}):

$$P = V_{DD} \times I_{DD}$$

Power Consumption $\approx 149.877 \text{ uW}$, of Differential Pair

Power Consumption $\approx 660.117 \text{ uW}$, of Folded Cascode

The differential amplifier-based BGR consumes less power due to its simpler structure and lower bias current requirements. The folded cascode design, while offering better performance, consumes higher power due to additional biasing branches and increased circuit complexity.

D. Power Supply Rejection Ratio (PSRR)

AC analysis was performed to evaluate supply noise rejection.

The folded cascode design achieves better PSRR performance, especially at low frequencies, due to cascoding which isolates the output from supply noise. The differential amplifier shows lower rejection due to limited gain.

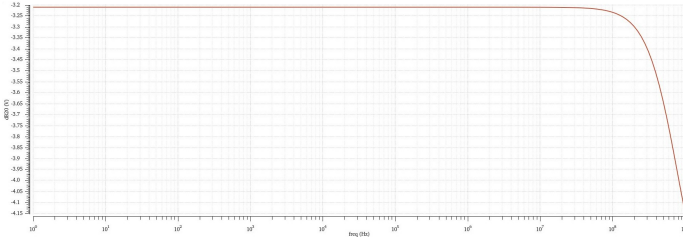


Fig. 8. PSRR vs frequency for differential amplifier-based BGR.

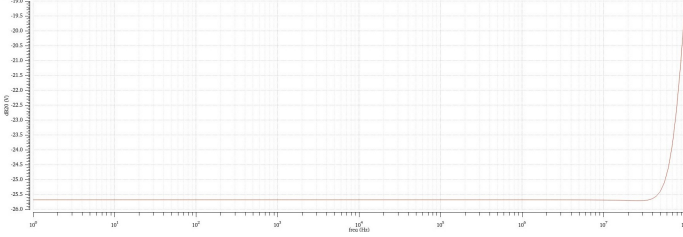


Fig. 9. PSRR vs frequency for folded cascode-based BGR.

E. Load Regulation Analysis

The load current was varied to observe output stability. The folded cascode topology maintains a more stable output under varying load conditions due to higher output resistance. The differential design shows slightly higher variation in V_{ref} .

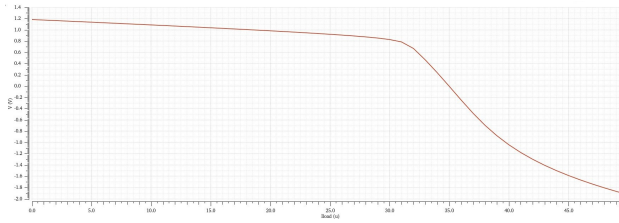


Fig. 10. Load regulation for differential amplifier-based BGR.

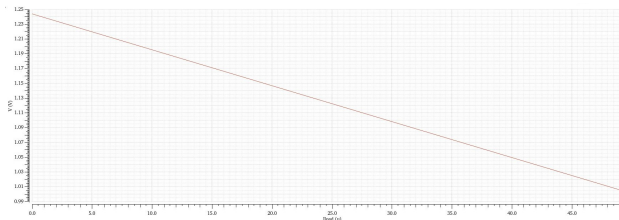


Fig. 11. Load regulation for folded cascode-based BGR.

F. Performance Comparison

Across all performance metrics, the folded cascode amplifier-based BGR consistently outperforms the differential amplifier-based design in terms of temperature stability, PSRR, line regulation, and load regulation. However, the differential amplifier offers lower power consumption and simpler implementation, making it suitable for low-power applications where ultra-high precision is not critical.

A comparative summary of the two designs is presented in Table I.

TABLE I
PERFORMANCE COMPARISON OF BGR DESIGNS

Parameter	Differential	Folded Cascode
Temperature Coefficient	Higher	Lower
PSRR	Moderate	High
Line Regulation	Moderate	Better
Power Consumption	Low	Moderate
Load Regulation	Moderate	Better

V. CONCLUSION

In this paper, two BGR circuits based on diodes have been proposed and compared under a 180nm CMOS technology. The study has been performed to examine the effect of different feedback amplifier topologies by using a two-stage NMOS differential amplifier and a folded cascode amplifier in a BGR circuit with the same core structure.

From simulation results, it is evident that the folded cascode-based design offers improved characteristics in terms of temperature coefficient, PSRR, line regulation, and load regulation. These performances are mainly due to the higher open-loop gain and the output impedance, leading to the reduction of input-referred offset voltage and supply voltage variation sensitivity.

On the other hand, the two-stage differential amplifier-based BGR shows relatively lower power dissipation and simpler design structure, which makes it feasible for low-power applications. Nevertheless, the lesser gain provided by the differential amplifier topology contributes to lower accuracy and environmental sensitivity.

Finally, the outcomes prove the essential role played by the amplifier topology in determining BGR performance.

VI. FUTURE SCOPE

In the current work, the design and simulation comparison of diode based BGR circuit utilizing two amplifier architectures has been done. There exist several possibilities to extend this study, making it even more convincing:

First, both circuits could be implemented using physical layouts, and post-layout simulations should be performed to investigate the influence of parasitics and transistor mismatches on such key figures as the temperature coefficient and PSRR.

Moreover, optimization of the size and biasing of transistors may be performed with respect to power dissipation and performance, especially for the folded cascode circuit.

It is also worth to conduct the experimental validation of simulations, implementing the proposed circuits as hardware or fabricated in silicon.

Another way to extend the work would be to evaluate the performance of the circuits in different process corners (TT, SS, FF).

Lastly, the comparative study should be expanded with other types of amplifier architecture.

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